

```
global_data = {
```

```
# — ATTRIBUTE IDENTIFI
```

```
"attribute_1": "total_cost",  
"attribute_2": "material_continuity",  
"attribute_3": "conductor_size_mm2",  
"attribute_4": "cable_length_m",  
"attribute_5": "voltage_class_kv",
```

```
# — DIRECTIONS
```

```
"direction_attribute_1": "negative",  
"direction_attribute_2": "positive",  
"direction_attribute_3": "positive",  
"direction_attribute_4": "positive",  
"direction_attribute_5": "positive",
```

```
# — ATTR
```

```
# ATTRIBUTE 1: total_cost (USD)
```

```
# OLD: 7,000 – 30,000
```

```
# NEW: 500 – 1,000,000
```

```
# Floor ($500): ~50 m of 16 mm2 Al cable in India — absolute minimum
```

```
# credible URD project (short residential service entrance)
```

```
# Ceiling ($1M): ~8,000 m of 500 mm2 Cu, full neutral, European labor rate —
```

```
# a large-scale primary feeder replacement; physically possible
```

```
# within the bounds of the other attributes
```

```
"attribute_1_min": 500,
```

```
"attribute_1_max": 1000000,
```

```
# ATTRIBUTE 2: material_continuity (binary 0 or 1)
```

```
# No change — 0 and 1 are the absolute mathematical limits of a binary attribute.
```

```
# No exaggeration possible or needed.
```

```
"attribute_2_min": 0,
```

```
"attribute_2_max": 1,
```

```
# ATTRIBUTE 3: conductor_size_mm2 (cross-sectional area)
```

```
# OLD: 35 – 500 mm2
```

```
# NEW: 10 – 1200 mm2
```

```
# Floor (10 mm2): Below standard URD, but covers any edge-case small
```

```

#           service drop or pilot conductor ever specified under
#           an IEC or AWG regime
# Ceiling (1200 mm²): Largest IEC-standard MV distribution cable conductor
#           commercially manufactured (e.g., 1200 mm² XLPE cables
#           used in dense urban networks in Germany/UK);
#           one step above the 1000 mm² max in common use
"attribute_3_min": 10,
"attribute_3_max": 1200,

# ATTRIBUTE 4: cable_length_m (project run length in meters)
# OLD: 100 – 10,000 m
# NEW: 25 – 25,000 m
# Floor (25 m): Absolute shortest credible URD project — a pad-mounted
#           transformer stub to a meter base on a commercial lot
# Ceiling (25,000 m = 25 km): Covers extended rural primary distribution
#           feeders in India or China electrification programs;
#           longer than typical but physically used in RDSS/DDUGJY
#           project procurement batches
"attribute_4_min": 25,
"attribute_4_max": 25000,

# ATTRIBUTE 5: voltage_class_kV
# OLD: 5 – 35 kV
# NEW: 5 – 46 kV
# Floor (5 kV): Unchanged — lowest recognized medium-voltage
#           distribution class globally
# Ceiling (46 kV): Prysmian EmPowr Link URD cable is explicitly rated
#           15–46 kV; some North American utilities use 34.5 kV
#           distribution systems, and 46 kV is the next recognized
#           ANSI distribution voltage class above 35 kV —
#           a modest but meaningful extension
"attribute_5_min": 5,
"attribute_5_max": 46,

# — COMMODITY PRICES (unchan
"copper_price_per_ton": 9500,
"aluminum_price_per_ton": 2500,
"copper_price_per_kg": 9.500, # fixed global Cu price
"aluminum_price_per_kg": 2.500, # fixed global Al price
}

```

```

regional_data = [

    {
        "region": "United States",
        # US primary URD is copper-dominant
        "copper_product_market_share": 0.65,
        "aluminum_product_market_share": 0.35,
        # Weights (established prior session — sum = 1.00)
        "weight_attribute_1": 0.30, # total_cost
        "weight_attribute_2": 0.25, # material_continuity
        "weight_attribute_3": 0.20, # conductor_size
        "weight_attribute_4": 0.15, # cable_length
        "weight_attribute_5": 0.10, # voltage_class
    },

    {
        "region": "China",
        # China 10 kV distribution increasingly aluminum-dominant (state-directed shift)
        "copper_product_market_share": 0.40,
        "aluminum_product_market_share": 0.60,
        # Weights (established prior session — sum = 1.00)
        "weight_attribute_1": 0.40, # total_cost — SOE tender lowest-bid culture
        "weight_attribute_2": 0.15, # material_continuity — actively transition
        "weight_attribute_3": 0.20, # conductor_size
        "weight_attribute_4": 0.20, # cable_length — large-scale urbanization multiplier
        "weight_attribute_5": 0.05, # voltage_class — near-standardized at 10 kV
    },

    {
        "region": "India",
        # India distribution is strongly aluminum due to DISCOM budget constraints
        "copper_product_market_share": 0.25,
        "aluminum_product_market_share": 0.75,
        # Weights (established prior session — sum = 1.00)
        "weight_attribute_1": 0.45, # total_cost — dominant driver in
        "weight_attribute_2": 0.20, # material_continuity — aluminum-directional inertia
        "weight_attribute_3": 0.15, # conductor_size
        "weight_attribute_4": 0.15, # cable_length
        "weight_attribute_5": 0.05, # voltage_class — standardized at 11 kV nationally
    }
]

```

```

    },
    {
      "region": "Europe",
      # Europe MV underground is copper-dominant, especially Western/Northern EU
      "copper_product_market_share": 0.70,
      "aluminum_product_market_share": 0.30,
      # Weights (established prior session — sum = 1.00)
      "weight_attribute_1": 0.25, # total_cost — lifecycle cost evaluated, not just upfront
      "weight_attribute_2": 0.25, # material_continuity — strong copper lega
      "weight_attribute_3": 0.20, # conductor_size — conduit space is con
      "weight_attribute_4": 0.15, # cable_length
      "weight_attribute_5": 0.15, # voltage_class — HIGHEST weight: 10
    },
  ]

```

product_data = [

```

{
  "region": "United States",
  "dominant material": "cu",
  "product": "LLDPE Jacket, Full Cu Concentric Neutral, URD Cable",
  #
  # Mass breakdown (per Prysmian EmPowr Link spec, scaled to 500 m):
  # Cu conductor: 485 kg/km × 0.5 km = 242.5 kg
  # Cu neutral: 515 kg/km × 0.5 km = 257.5 kg → total Cu = 500 kg
  # Non-metallic: 398 kg/km × 0.5 km = 199 kg → rounds to 200 kg
  "copper_kg": 500, # conductor + concentric neutral
  "aluminum_kg": 0,
  "total_mass_kg": 700, # 500 Cu + 200 non-metallic
  #
  # Non-material cost ($12,000/km × 0.5 km = $6,000)
  # Covers: XLPE insulation, LLDPE jacket, semi-con shields,
  # manufacturing labor, overhead, logistics (US rate)
  "non_material_cost_per_unit": 6000,
  #
  # total_cost = 500 × 9.50 + 0 × 2.50 + 6,000 = $10,750
  #

```

```

"attribute_2_value": 1, # US is copper-history dominant → Cu product
"attribute_3_value": 53.5, # 1/0 AWG = 53.5 mm² (canonical from prior session)
"attribute_4_value": 500, # 500 m run
"attribute_5_value": 15, # 15 kV class (12.47 kV GrdY system)
},

{
  "region": "United States",
  "dominant material": "al",
  "product": "LLDPE Jacket, Full Cu Concentric Neutral, URD Cable",
  #
  # Mass breakdown (scaled to 500 m):
  # Al conductor: 182 kg/km × 0.5 km = 91 kg (67.4 mm² × 2.7 g/cm³)
  # Cu neutral: 628 kg/km × 0.5 km = 314 kg → rounded to 315 kg
  # (matches Prysmian 2/0 neutral count; neutral is Cu regard
  # Non-metallic: 448 kg/km × 0.5 km = 224 kg
  "copper_kg": 315, # Cu concentric neutral only (conductor is Al)
  "aluminum_kg": 91, # Al conductor only
  "total_mass_kg": 630, # 315 + 91 + 224 non-metallic
  #
  "non_material_cost_per_unit": 6000, # same rate as Cu
  #
  # total_cost = 315 × 9.50 + 91 × 2.50 + 6,000
  # = 2,992.50 + 227.50 + 6,000 = $9,220
  #
  "attribute_2_value": 0, # US copper history → Al product = no continuity
  "attribute_3_value": 67.4, # 2/0 AWG = 67.4 mm² (ampacity match to 1/0 Cu)
  "attribute_4_value": 500,
  "attribute_5_value": 15,
},

{
  "region": "China",
  "dominant material": "cu",
  "product": "LLDPE Jacket, 1/3 Cu Concentric Neutral, URD Cable",
  #
  # Mass breakdown (per km, scaled to 2,000 m):
  # Cu conductor: 70 mm² × 8.9 g/cm³ = 623 kg/km → 620 kg/km × 2 = 1,240 kg
  # Cu neutral (1/3 × 70 mm² = 23.3 mm², 7× #12 AWG wires):
  # 7 × 29.5 kg/km = 206 kg/km → 205 kg/km × 2 = 410 kg

```

```

# Total Cu:    1,240 + 410 = 1,650 kg
# Non-metallic (10 kV thinner insulation): 275 kg/km × 2 = 550 kg
"copper_kg":    1650,
"aluminum_kg":  0,
"total_mass_kg": 2200, # 1,650 + 550 non-metallic
#
"non_material_cost_per_unit": 12000, # $6,000/km × 2 km (China la
#
# total_cost = 1,650 × 9.50 + 0 × 2.50 + 12,000
#           = 15,675 + 12,000 = $27,675
#
"attribute_2_value": 0, # China is transitioning to Al → Cu has no
"attribute_3_value": 70, # 70 mm² (standard IEC MV distribution size in China)
"attribute_4_value": 2000,
"attribute_5_value": 10, # 10 kV (national standard for Chinese distribution grid)
},

{
  "region": "China",
  "dominant material": "al",
  "product": "LLDPE Jacket, 1/3 Cu Concentric Neutral, URD Cable",
  #
  # Mass breakdown (per km, scaled to 2,000 m):
  # Al conductor: 95 mm² × 2.7 g/cm³ = 257 kg/km → 255 kg/km × 2 = 510 kg
  # Cu neutral (1/3 × 95 mm² = 31.7 mm², 10× #12 AWG wires):
  #           10 × 29.5 kg/km = 295 kg/km × 2 = 590 kg
  # Non-metallic: 300 kg/km × 2 = 600 kg
  "copper_kg":    590, # Cu concentric neutral only
  "aluminum_kg":  510, # Al conductor only
  "total_mass_kg": 1700, # 590 + 510 + 600 non-metallic
  #
  "non_material_cost_per_unit": 12000,
  #
  # total_cost = 590 × 9.50 + 510 × 2.50 + 12,000
  #           = 5,605 + 1,275 + 12,000 = $18,880
  #
  "attribute_2_value": 1, # China is Al-transitioning → Al product = full continuity
  "attribute_3_value": 95, # 95 mm² Al ≈ ampacity match to 70
  "attribute_4_value": 2000,
  "attribute_5_value": 10,

```

},

```
{
  "region": "India",
  "dominant material": "cu",
  "product": "LLDPE Jacket, 1/3 Cu Concentric Neutral, URD Cable",
  #
  # Mass breakdown (same per-km basis as China Cu; 11 kV ≈
  # Cu conductor: 620 kg/km × 1.0 = 620 kg
  # Cu neutral: 205 kg/km × 1.0 = 205 kg → total Cu = 825 kg
  # Non-metallic: 275 kg/km × 1.0 = 275 kg
  "copper_kg": 825,
  "aluminum_kg": 0,
  "total_mass_kg": 1100, # 825 + 275 non-metallic
  #
  "non_material_cost_per_unit": 5000, # $5,000/km (India — I
  #
  # total_cost = 825 × 9.50 + 0 × 2.50 + 5,000
  #           = 7,837.50 + 5,000 = $12,837.50
  #
  "attribute_2_value": 0, # India is Al-dominant → Cu has no material continuity
  "attribute_3_value": 70, # 70 mm² (same IEC standard as China)
  "attribute_4_value": 1000,
  "attribute_5_value": 11, # 11 kV (British colonial legacy, national standard)
},
```

```
{
  "region": "India",
  "dominant material": "al",
  "product": "LLDPE Jacket, 1/3 Cu Concentric Neutral, URD Cable",
  #
  # Mass breakdown (same per-km basis as China Al):
  # Al conductor: 255 kg/km × 1.0 = 255 kg
  # Cu neutral: 295 kg/km × 1.0 = 295 kg
  # Non-metallic: 300 kg/km × 1.0 = 300 kg
  "copper_kg": 295,
  "aluminum_kg": 255,
  "total_mass_kg": 850, # 295 + 255 + 300 non-metallic
  #
```

```

"non_material_cost_per_unit": 5000,
#
# total_cost = 295 × 9.50 + 255 × 2.50 + 5,000
#           = 2,802.50 + 637.50 + 5,000 = $8,440
#
"attribute_2_value": 1, # India is Al-dominant → Al product = full continuity
"attribute_3_value": 95, # 95 mm² Al
"attribute_4_value": 1000,
"attribute_5_value": 11,
},

```

```

{
  "region": "Europe",
  "dominant material": "cu",
  "product": "LLDPE Jacket, 1/3 Cu Concentric Neutral, URD Cable",
  #
  # Mass breakdown (per km, scaled to 800 m):
  # Cu conductor: 95 mm² × 8.9 g/cm³ = 845 kg/km × 0.8 = 676 kg
  # Cu neutral (1/3 × 95 mm² = 31.7 mm², 10× #12 AWG wires):
  #           295 kg/km × 0.8 = 236 kg
  # Total Cu: 676 + 236 = 912 kg
  # Non-metallic (20 kV, thicker insulation ~5.5 mm wall):
  #           485 kg/km × 0.8 = 388 kg
  "copper_kg": 912,
  "aluminum_kg": 0,
  "total_mass_kg": 1300, # 912 + 388 non-metallic
  #
  "non_material_cost_per_unit": 11200, # $14,000/km × 0.8 km (Eur
  #
  # total_cost = 912 × 9.50 + 0 × 2.50 + 11,200
  #           = 8,664 + 11,200 = $19,864
  #
  "attribute_2_value": 1, # Europe is copper-dominant → Cu product
  "attribute_3_value": 95, # 95 mm² (larger than Asia due to 2
  "attribute_4_value": 800,
  "attribute_5_value": 20, # 20 kV (most common new-build
},

```

```

{

```



```

"region": "Europe",
"dominant material": "al",
"product": "LLDPE Jacket, 1/3 Cu Concentric Neutral, URD Cable",
#
# Mass breakdown (per km, scaled to 800 m):
# Al conductor:  $120 \text{ mm}^2 \times 2.7 \text{ g/cm}^3 = 324 \text{ kg/km} \times 0.8 = 259 \text{ kg}$ 
# Cu neutral ( $1/3 \times 120 \text{ mm}^2 = 40 \text{ mm}^2$ , 12× #12 AWG wires):
#  $12 \times 29.5 = 354 \text{ kg/km} \times 0.8 = 283 \text{ kg}$ 
# Non-metallic (20 kV, slightly more than Cu due to larger OD):
#  $501 \text{ kg/km} \times 0.8 = 401 \text{ kg}$ 
"copper_kg": 283, # Cu concentric neutral only
"aluminum_kg": 259, # Al conductor only
"total_mass_kg": 943, # 283 + 259 + 401 non-metallic
#
"non_material_cost_per_unit": 11200,
#
# total_cost =  $283 \times 9.50 + 259 \times 2.50 + 11,200$ 
#  $= 2,688.50 + 647.50 + 11,200 = \$14,536$ 
#
"attribute_2_value": 0, # Europe copper-dominant → Al product = no continuity
"attribute_3_value": 120, # 120 mm² Al ≈ ampacity match t
"attribute_4_value": 800,
"attribute_5_value": 20,
},
]

```